Development Summary - Outfit Recommendation System

1. Purpose of this document

The purpose of this document is to summarize the development process of the Outfit Recommendation System project.

This document explains the main experiments, the results, the development decisions, and the improvements made throughout the project. It also shows how the project evolved from simple classification baselines into a working recommendation prototype based on style prediction, clothing type prediction, and embedding-based retrieval.

The document is intended to support the final project documentation by providing a clear overview of what was built, why certain decisions were made, and what the main findings were.

2. Development overview

The project was developed iteratively. The original goal was to build an outfit recommendation system, but direct outfit compatibility was too subjective and difficult to model immediately. Because of this, the first development step focused on two simpler classification tasks: fashion style classification and clothing type classification.

After both classifiers were built, they were connected to a basic rule-based recommendation prototype. This prototype recommended items with the same predicted style and a different predicted clothing type. The next prototype improved this by using image embeddings and cosine similarity to rank recommendations by visual similarity.

After the first prototypes worked on the curated dataset, a real-world evaluation set was created. This showed that the style classifier struggled with realistic images, while the clothing type classifier remained more reliable. Because style prediction directly affects recommendation quality, the next development iterations focused on improving the style classifier.

A partial fine-tuning experiment improved curated test accuracy but made real-world performance worse. This showed that model changes alone were not enough. A later dataset-improvement experiment added targeted extra training images, especially for streetwear. This improved real-world style accuracy from 0.4750 to 0.6000.

The improved style model was then tested inside the embedding-based recommendation pipeline. The results showed that recommendations became more useful when style prediction was correct, but the system was still affected by style and clothing type prediction errors.

Because some recommendation failures were caused by wrong clothing type predictions, the next development step focused on improving the type classifier with targeted real-world-like images. This improved real-world type accuracy from 0.7500 to 0.8875.

The improved type model was then tested inside the full recommendation pipeline. The results showed that type-related structural recommendation failures were reduced. In the selected real-world examples, same-

actual-type recommendation issues dropped from 2 examples to 0 examples. However, the system still had remaining style-related weaknesses, especially when gothic or sporty items were predicted as streetwear.

After this, a more systematic recommendation evaluation framework was created. Instead of only inspecting a few recommendation examples visually, 32 balanced real-world examples were evaluated using automatic checks and manual scoring criteria. This showed that the recommender was structurally strong, with an average structural validity score of 1.875 out of 2, but weaker in style consistency, visual coherence, and overall recommendation quality, each averaging 1.3125 out of 2. This confirmed that the next major improvement should focus on style reliability.

3. Experiment 1: Style Classification Baseline

The first experiment focused on fashion style classification.

The goal was to test whether the manually collected dataset could be used to train a model that predicts the visual style of an individual clothing item.

The four style classes were:

- formal
- gothic
- sporty
- streetwear

A pretrained ResNet34 model was used with transfer learning. The pretrained feature extractor was frozen, and only the final classification layer was trained.

The model was trained on the curated style split:

- 560 training images
- 120 validation images
- 120 test images

The style baseline achieved a curated test accuracy of **0.8417** and a macro F1-score of **0.84**.

The results showed that style classification was feasible, but some style categories were harder than others. Gothic and formal performed strongly, while streetwear was the weakest class. This suggested that streetwear visually overlaps more with other styles, especially sporty and gothic.

4. Experiment 2: Clothing Type Classification Baseline

The second experiment focused on clothing type classification.

The goal was to test whether the model could identify what kind of clothing item was shown in an image. This was important because the recommendation system needs to avoid recommending the same type of item as the input.

The four clothing type classes were:

- jacket
- pants
- shoes

- tshirt

The same ResNet34 transfer learning approach was used. The pretrained feature extractor was frozen, and only the final classification layer was trained.

The clothing type classifier achieved an internal test accuracy of **1.00** on the curated test split. Because this result was unusually high, a small external test set of 20 images was created. On this external test set, the model achieved **0.90** accuracy.

Later, during the larger real-world evaluation, the clothing type classifier achieved **0.7500** accuracy.

These results showed that clothing type classification was more reliable than style classification. This makes sense because clothing types have clearer visual structure than fashion styles.

5. Experiment 3: Rule-Based Recommendation Prototype

The third experiment connected the style and clothing type classifiers to a first recommendation prototype.

The goal was to prove that the two classification models could be used together in a basic recommendation workflow.

The rule-based recommender used the following logic:

```
Recommend items where:
    item style = predicted style
    item type != predicted type
```

For example, if the input image was predicted as a gothic jacket, the system recommended gothic items that were not jackets, such as gothic pants, gothic shoes, and gothic tshirts.

This prototype showed that the classification outputs could be used to generate structured outfit suggestions. However, the recommended items were selected randomly from the filtered catalogue. This meant that the system could produce logically valid recommendations, but it did not yet rank items by visual compatibility.

6. Experiment 4: Embedding-Based Recommendation Prototype

The fourth experiment improved the rule-based recommendation prototype by adding visual similarity.

Instead of randomly selecting items from the filtered catalogue, the system used image embeddings extracted from a pretrained ResNet34 feature extractor. These embeddings represented each image as a feature vector.

The recommendation process became:

1. Predict the style of the input image.
2. Predict the clothing type of the input image.
3. Filter the catalogue to items with the same predicted style.
4. Exclude items with the same predicted clothing type.
5. Compute cosine similarity between the input image embedding and candidate embeddings.
6. Select the most similar item from each remaining clothing type.

This improved the recommendation system because items were no longer selected randomly. The system could now rank candidates based on visual similarity, making the recommendations more structured and explainable.

This experiment showed that embedding-based retrieval was a stronger foundation for recommendation than simple rule-based random selection.

7. Experiment 5: Real-World Evaluation

After the baseline models and recommendation prototypes were created, a separate real-world test set was collected.

The purpose of this evaluation was to test whether the trained models could generalize beyond the clean curated dataset.

The real-world test set contained **80 images**:

- 4 styles
- 4 clothing types
- 5 images per style/type combination

The images were more realistic than the original dataset and included more varied presentation styles, such as worn clothing, different backgrounds, and less controlled image conditions.

The results were:

Model	Real-world accuracy
Style classifier	0.4750
Clothing type classifier	0.7500

The results showed that the clothing type classifier generalized better than the style classifier. The biggest issue was the style classifier, especially the streetwear class. Streetwear had **0.00 recall**, meaning that none of the real-world streetwear images were correctly predicted as streetwear.

This evaluation was important because it revealed that good curated test performance was not enough. The system needed to be tested on more realistic images before improving the recommendation pipeline further.

8. Experiment 6: Style Classifier Fine-Tuning

After the real-world evaluation showed weak style generalization, the next experiment focused on improving the style classifier.

The original style classifier trained only the final classification layer. In this experiment, the last ResNet34 block, `layer4`, was unfrozen together with the final classification layer. This allowed the model to adapt more to fashion-specific visual features.

The fine-tuned model achieved:

Evaluation set	Accuracy
Curated test set	0.9417
Real-world test set	0.4250

The curated test accuracy improved strongly compared with the original style baseline, which achieved **0.8417**. However, real-world accuracy decreased from **0.4750** to **0.4250**.

This showed that fine-tuning helped the model perform better on clean curated images, but it did not improve generalization to realistic images. The model likely adapted too strongly to the curated dataset.

This experiment was useful because it showed that changing the training strategy alone was not enough. The main problem was likely the mismatch between the curated training data and the real-world test images.

9. Experiment 7: Style Dataset Improvement

After the fine-tuning experiment failed to improve real-world performance, the next experiment focused on improving the style training dataset.

The real-world evaluation showed that the main problem was the style classifier, especially streetwear. Because of this, extra targeted training images were collected and added to a new dataset folder called `style_extra`.

The extra dataset contained **100 images**:

- 60 streetwear images
- 20 formal images
- 20 sporty images

The images were selected based on the main failure cases from the real-world evaluation:

- streetwear was often confused with gothic or sporty
- formal pants and shoes were often misclassified
- sporty jackets and shoes were often confused with other styles

The new model was trained using the original style training set plus the extra targeted images. The model used the same basic training strategy as the original baseline, where the pretrained ResNet34 feature extractor was frozen and only the final classification layer was trained.

The dataset-improved model achieved:

Evaluation set	Accuracy
Curated test set	0.8667
Real-world test set	0.6000

This was the best real-world style result so far.

The most important improvement was streetwear recall. In the original real-world evaluation, streetwear recall was **0.00**. In the dataset-improved model, streetwear recall increased to **0.90**.

However, the model also started predicting streetwear too often. Many gothic and sporty images were incorrectly classified as streetwear. This means that the extra data helped the model recognize streetwear, but also shifted the model too strongly toward the streetwear class.

Overall, this experiment showed that dataset improvement was more effective than fine-tuning alone for improving real-world generalization.

10. Experiment 8: Recommendation with Improved Style Model

The eighth experiment tested the embedding-based recommendation system using the improved style classifier.

The recommendation system used:

- the dataset-improved style classifier
- the existing clothing type classifier
- the cleaned catalogue dataset
- ResNet34 image embeddings
- cosine similarity ranking

The goal was to check whether the improved style classifier helped the recommender produce better results for real-world input images.

Several real-world examples were tested. The results showed that the improved style model helped when the style prediction was correct. For example, the system produced useful recommendations for examples such as formal shoes, gothic jackets, sporty jackets, and streetwear tshirts.

However, the recommendation pipeline was still affected by classification errors. If the style prediction was wrong, the recommender retrieved items from the wrong style category. If the clothing type prediction was wrong, the system sometimes recommended another item from the same actual clothing category.

For example, a streetwear jacket was correctly predicted as streetwear, but incorrectly predicted as a tshirt. Because the recommender excluded tshirts instead of jackets, it recommended another streetwear jacket. This showed that recommendation quality depends on both the style classifier and the clothing type classifier.

This experiment confirmed that the improved style model was useful, but the full recommendation system still needs stronger safeguards and more robust prediction handling.

11. Experiment 9: Recommendation with Confidence Safeguards

The ninth experiment improved the recommendation pipeline by adding confidence-based safeguards.

The previous recommendation experiment showed that the recommender still depended strongly on the correctness of the style and clothing type classifiers. When the predicted type was wrong, the system could exclude the wrong clothing category and recommend another item of the same actual type. When the predicted style was wrong, the system retrieved items from the wrong style category.

To make the system more transparent, confidence thresholds were added:

- style confidence threshold: 0.60
- type confidence threshold: 0.60

The system assigns one of four statuses:

- reliable
- type_uncertain
- style_uncertain
- style_and_type_uncertain

The recommender still generates recommendations, but it now adds warnings when the style or clothing type confidence is low.

On the eight selected real-world examples, the safeguard results were:

Status	Count
reliable	4
type_uncertain	2
style_uncertain	2
style_and_type_uncertain	0

This showed that confidence safeguards can make some uncertain recommendation cases more explainable. For example, the streetwear jacket example was correctly marked as `type_uncertain` because the type classifier predicted `tshirt` with low confidence.

However, the experiment also showed that confidence thresholds are not a complete reliability solution. The gothic tshirt example was incorrectly predicted as streetwear with high confidence, so it was still marked as reliable. This shows that the model can be confidently wrong.

Overall, this experiment improved the transparency of the recommendation system, but future work would need better calibration or fallback behavior for uncertain predictions.

12. Experiment 10: Clothing Type Dataset Improvement

The tenth experiment focused on improving the clothing type classifier using targeted real-world-like training data.

Previous recommendation experiments showed that type prediction errors could break the recommendation structure. If the input type was predicted incorrectly, the recommender excluded the wrong category and could recommend another item of the same actual type.

For example, a streetwear jacket was correctly predicted as streetwear, but incorrectly predicted as a tshirt. Because the recommender excluded tshirts instead of jackets, it recommended another streetwear jacket. This showed that type prediction was not only a classifier issue, but a structural issue for the recommendation system.

To address this, a new dataset folder called `type_extra` was created. It contained 90 targeted extra images:

- 30 jacket images
- 30 tshirt images
- 15 pants images

- 15 shoes images

The extra images focused on real-world-like cases, especially worn jackets, open jackets, oversized tshirts, cropped upper-body views, pants, and shoes in more varied presentation styles.

The improved type classifier was trained using the original type training split plus the `type_extra` dataset. The model architecture and training setup stayed the same as the original type baseline:

- pretrained ResNet34
- frozen feature extractor
- replaced final classification layer
- trained final layer only

This made the experiment focused on the effect of dataset improvement rather than architecture changes.

The improved type model achieved:

Evaluation set	Accuracy
Curated type test set	1.0000
External type test set	0.9000
Real-world type test set	0.8875

The curated and external test results stayed the same as the original type model, meaning that the extra data did not damage performance on cleaner evaluation sets.

The most important improvement was on the real-world test set. The original type classifier achieved 0.7500 real-world accuracy, while the improved type classifier achieved 0.8875. This showed that targeted type data helped the model generalize better to realistic input images.

The remaining issue was that some pants and tshirts were still predicted as jackets. This suggests that the added jacket data helped the model recognize jackets better, but also made the jacket class slightly more dominant in ambiguous cases.

Overall, this experiment showed that improving the type dataset directly improved the model where the recommendation pipeline needed it most.

13. Experiment 11: Recommendation with Improved Models

The eleventh experiment tested the full recommendation pipeline using both improved classifiers.

The recommendation system used:

- the dataset-improved style classifier
- the dataset-improved type classifier
- the cleaned catalogue dataset
- ResNet34 image embeddings
- cosine similarity ranking
- confidence safeguards

The goal was to check whether the improved type classifier reduced structural recommendation failures in the full pipeline.

The experiment compared two setups:

Setup	Style model	Type model
Previous setup	`style_resnet34_extra_data.pth`	`type_resnet34.pth`
Improved setup	`style_resnet34_extra_data.pth`	`type_resnet34_extra_data.pth`

The same eight real-world examples from the confidence safeguard experiment were tested again. This made the comparison fair because the input images stayed the same.

The improved type model fixed two type prediction errors:

- a formal jacket was previously predicted as pants, but was now correctly predicted as jacket
- a streetwear jacket was previously predicted as tshirt, but was now correctly predicted as jacket

The type prediction comparison was:

Metric	Previous setup	Improved setup
Correct type predictions	6 / 8	8 / 8
Type errors fixed	N/A	2
Type errors introduced	N/A	0
Reliable examples	4 / 8	6 / 8

The recommendation structure also improved. With the previous setup, 2 out of 8 examples included a recommendation with the same actual type as the input item. With the improved setup, this dropped to 0 out of 8.

Metric	Previous setup	Improved setup
Same-actual-type recommendation issues	2	0
Structural issues fixed	N/A	2
Structural issues introduced	N/A	0

This showed that the improved type classifier did not only improve standalone classification accuracy. It also improved the recommendation pipeline by preventing duplicate actual clothing types in the output.

However, the experiment also showed that some remaining issues are style-related. For example, some gothic and sporty images were still predicted as streetwear. In these cases, the recommendation structure can be correct, but the retrieved items may still come from the wrong style category.

Overall, this experiment confirmed that the improved type classifier is the better choice for the current prototype. It reduced type-related structural failures, while the main remaining bottleneck became style reliability and recommendation evaluation.

14. Experiment 12: Recommendation Evaluation Framework

The twelfth experiment focused on evaluating the current final recommendation prototype more systematically.

Previous recommendation experiments used selected examples and visual inspection. This was useful for understanding the system qualitatively, but it was not repeatable enough to compare future improvements. Because of this, a small evaluation framework was created.

The evaluation used the current final prototype:

- `style_resnet34_extra_data.pth`
- `type_resnet34_extra_data.pth`
- ResNet34 image embeddings
- cosine similarity ranking
- confidence safeguards

A balanced subset of the real-world test set was selected:

- 4 styles
- 4 clothing types
- 2 examples per style/type combination
- 32 examples in total

For each example, the system recorded automatic information such as:

- true style
- true clothing type
- predicted style
- predicted clothing type
- style confidence
- type confidence
- safeguard status
- recommended item types
- whether the recommendations included the same actual type as the input

The automatic evaluation showed:

Metric	Result
Style prediction accuracy on selected examples	0.5625
Type prediction accuracy on selected examples	0.9375
Same-actual-type recommendation issues	2 / 32

This confirmed that clothing type prediction was now strong, while style prediction remained the main bottleneck.

A manual scoring framework was then added. Each recommendation was scored from 0 to 2 using the following criteria:

- structural validity
- style consistency
- visual coherence
- confidence warning usefulness
- overall recommendation quality

The manual evaluation results were:

Criterion	Average score
Structural validity	1.8750
Style consistency	1.3125
Visual coherence	1.3125
Confidence warning usefulness	1.4688
Overall quality	1.3125

The overall recommendation quality distribution was:

Quality	Count
Bad	7
Partially useful	8
Good	17

This means that 17 out of 32 recommendations were scored as good, while 15 were either partially useful or bad.

The by-style results showed that streetwear had the best average overall quality score at 1.625, while gothic had the weakest score at 1.125. This fits the earlier findings: the extra style data improved streetwear recognition, but gothic and sporty boundary cases are still difficult.

The by-type results showed that jacket inputs performed best, with an average overall quality score of 1.750. Shoe inputs performed worst, with an average overall quality score of 0.625. This does not mean that shoe type prediction was weak. Instead, shoe inputs often caused style prediction errors, which led the recommender to retrieve items from the wrong style category.

Overall, this experiment showed that the current recommender is structurally reliable, but still limited by style prediction quality. The recommendation logic can usually select useful clothing categories, but it still struggles when the predicted style is wrong.

15. Overall Results Comparison

The table below summarizes the main experiments and results from the project.

Experiment	Main goal	Curated/Internal Result	Real-world Result	Main conclusion
Style classification baseline	Predict fashion style	0.8417 test accuracy	0.4750 real-world accuracy	The model learned style patterns on clean data but struggled with realistic images.
Clothing type classification baseline	Predict clothing type	1.00 internal test accuracy	0.90 small external / 0.7500 real-world accuracy	Clothing type classification was more reliable than style classification, but still made real-world errors.
Rule-based recommendation	Recommend same-style, different-type items	Qualitative	Not tested on real-world images	The pipeline worked, but recommendations were randomly selected.
Embedding-based recommendation	Rank recommendations by visual similarity	Qualitative	Not initially tested on real-world images	Embeddings improved recommendation ranking compared with random selection.
Real-world evaluation	Test generalization	N/A	Style: 0.4750, Type: 0.7500	Style prediction was the main weakness, but type errors also affected the recommender.
Style fine-tuning	Improve style classifier using partial fine-tuning	0.9417 curated accuracy	0.4250 real-world accuracy	Fine-tuning improved clean-data performance but did not improve real-world generalization.
Style dataset improvement	Improve style classifier using extra targeted data	0.8667 curated accuracy	0.6000 real-world accuracy	Extra data improved real-world style performance and was more useful than fine-tuning alone.

Experiment	Main goal	Curated/Internal Result	Real-world Result	Main conclusion
Recommendation with improved style model	Test recommender with improved style classifier	Qualitative	Qualitative	Recommendations improved when predictions were correct, but errors in style/type still affected output.
Recommendation with confidence safeguards	Add warnings for uncertain predictions	Qualitative	4 reliable, 2 type_uncertain, 2 style_uncertain on 8 examples	Confidence warnings improved transparency, but high-confidence wrong predictions still occurred.
Type dataset improvement	Improve type classifier using extra targeted data	1.0000 curated / 0.9000 external	0.8875 real-world accuracy	Targeted type data improved real-world type prediction while preserving clean-data performance.
Recommendation with improved models	Test recommender with improved style and type models	Qualitative	Same-actual-type issues reduced from 2 to 0 on 8 examples	Improved type prediction reduced structural recommendation failures.
Recommendation evaluation framework	Evaluate final prototype systematically	Automatic + manual evaluation	17 good, 8 partially useful, 7 bad recommendations on 32 examples	The recommender is structurally strong, but style consistency and visual coherence remain weaker.

16. Selected Models for Current Prototype

Based on the experiments, the current prototype uses the following models:

Component	Selected model	Reason
Style classifier	`style_resnet34_extra_data.pth`	This model achieved the best real-world style accuracy, improving from 0.4750 to 0.6000.
Clothing type classifier	`type_resnet34_extra_data.pth`	This model improved real-world type accuracy from 0.7500 to 0.8875 while

Component	Selected model	Reason
		preserving 1.0000 curated accuracy and 0.9000 external accuracy.
Recommendation ranking	ResNet34 embedding feature extractor	This provides visual similarity ranking using cosine similarity and improves the recommender compared with random selection.
Confidence safeguards	Softmax confidence thresholds	These make uncertain recommendations more transparent, using 0.60 thresholds for style and type confidence.

The selected style model is not the model with the highest curated test accuracy. The fine-tuned layer4 model achieved the highest curated test accuracy at 0.9417, but it performed worse on the real-world test set with 0.4250 accuracy. Because the recommendation system is intended to work with realistic input images, the dataset-improved style model is selected instead.

The selected type model is now the dataset-improved type classifier. The original type model already performed strongly on clean data, but the improved type model achieved much better real-world type accuracy, increasing from 0.7500 to 0.8875.

The current recommendation prototype therefore uses the dataset-improved style classifier, the dataset-improved type classifier, embedding-based retrieval, and confidence safeguards.

17. Main Development Decisions

Several important development decisions were made during the project.

1. The project was scoped down from direct outfit compatibility to classification-supported recommendation because outfit compatibility is subjective and difficult to evaluate directly.
2. Style and clothing type were treated as intermediate representations for recommendation.
3. ResNet34 transfer learning was used because the dataset was manually curated and relatively small.
4. The first recommendation prototype was rule-based to prove the basic pipeline before adding more complex ranking.
5. Embedding-based retrieval was added to improve recommendation quality compared with random selection.
6. A real-world evaluation set was created because curated test results were not enough to prove generalization.
7. Fine-tuning was tested, but the result showed that better curated performance does not automatically mean better real-world performance.
8. Extra targeted data was added after the error analysis showed that streetwear was the weakest class.

9. The improved style model was integrated back into the recommender to test whether model improvement helped the full pipeline.
10. The type classifier was improved after recommendation examples showed that type errors caused structural failures in the output.
11. The improved type model was selected because it reduced same-actual-type recommendation issues from 2 to 0 on the selected real-world examples.
12. A manual recommendation evaluation framework was added because visual inspection alone was not repeatable enough to compare future improvements.
13. The evaluation results showed that the next improvement should focus on style boundary cases, especially gothic, sporty, and streetwear confusion.

18. Final Conclusion

The final system is a working prototype for fashion recommendation based on image classification and embedding-based retrieval.

The system can predict the style and clothing type of an input image, then recommend visually similar items from the same predicted style and different clothing categories.

The strongest style-related technical result was the dataset-improved style model, which increased real-world style accuracy from **0.4750** to **0.6000**. This showed that improving the dataset was more effective for real-world generalization than fine-tuning alone.

The strongest type-related technical result was the dataset-improved type model, which increased real-world type accuracy from **0.7500** to **0.8875**. This showed that targeted real-world-like type data can improve the classifier without damaging curated or external test performance.

The recommendation pipeline works best when both style and type predictions are correct. When the style prediction is wrong, the system retrieves items from the wrong style category. When the type prediction is wrong, the system may recommend an item from the same actual clothing category.

The improved type model reduced type-related structural failures. In the selected recommendation comparison, same-actual-type recommendation issues dropped from **2** to **0**. This means the recommendation structure became more reliable.

The final recommendation evaluation confirmed this. Across 32 balanced real-world examples, the average structural validity score was **1.875** out of 2. This shows that the recommender usually suggests useful complementary clothing types.

However, style consistency, visual coherence, and overall recommendation quality each averaged **1.3125** out of 2. Out of 32 recommendations, **17** were scored as good, **8** were partially useful, and **7** were bad. This means that the system is useful as a prototype, but not yet reliable enough as a complete outfit recommendation system.

The main remaining weakness is style reliability. Some recommendations are structurally correct but stylistically weak because the style classifier still confuses gothic, sporty, and streetwear. The system also does not yet evaluate color matching, texture, proportions, occasion, or user preference.

Overall, the project demonstrates that classification and embedding retrieval can support a basic fashion recommendation system. At the same time, it also shows the limits of this approach when style prediction is uncertain or wrong.

19. Next Steps

The next development steps are:

1. Add more balanced style boundary examples between streetwear, gothic, and sporty so the style classifier does not overpredict streetwear.
2. Create a new style improvement experiment focused on boundary cases, especially:
    - gothic but not streetwear
    - streetwear but not gothic
    - sporty but not streetwear
    - streetwear but not sporty
    - formal tshirts that should not be confused with streetwear
3. Re-train or improve the style classifier using the new boundary data.
4. Re-test the recommendation pipeline using the same 32-example evaluation framework from Experiment 12.
5. Compare the new recommendation scores with the current evaluation results:
    - structural validity: 1.875
    - style consistency: 1.3125
    - visual coherence: 1.3125
    - confidence warning usefulness: 1.4688
    - overall quality: 1.3125
6. Improve recommendation ranking with color features, fashion-specific compatibility features, or CLIP embeddings.
7. Add a simple user evaluation where people rate whether the recommended outfits make sense.
8. Expand the recommendation catalogue with more varied clothing items so the system has better options to retrieve from.
9. Keep the real-world test set separate from training data so future improvements can be evaluated fairly.